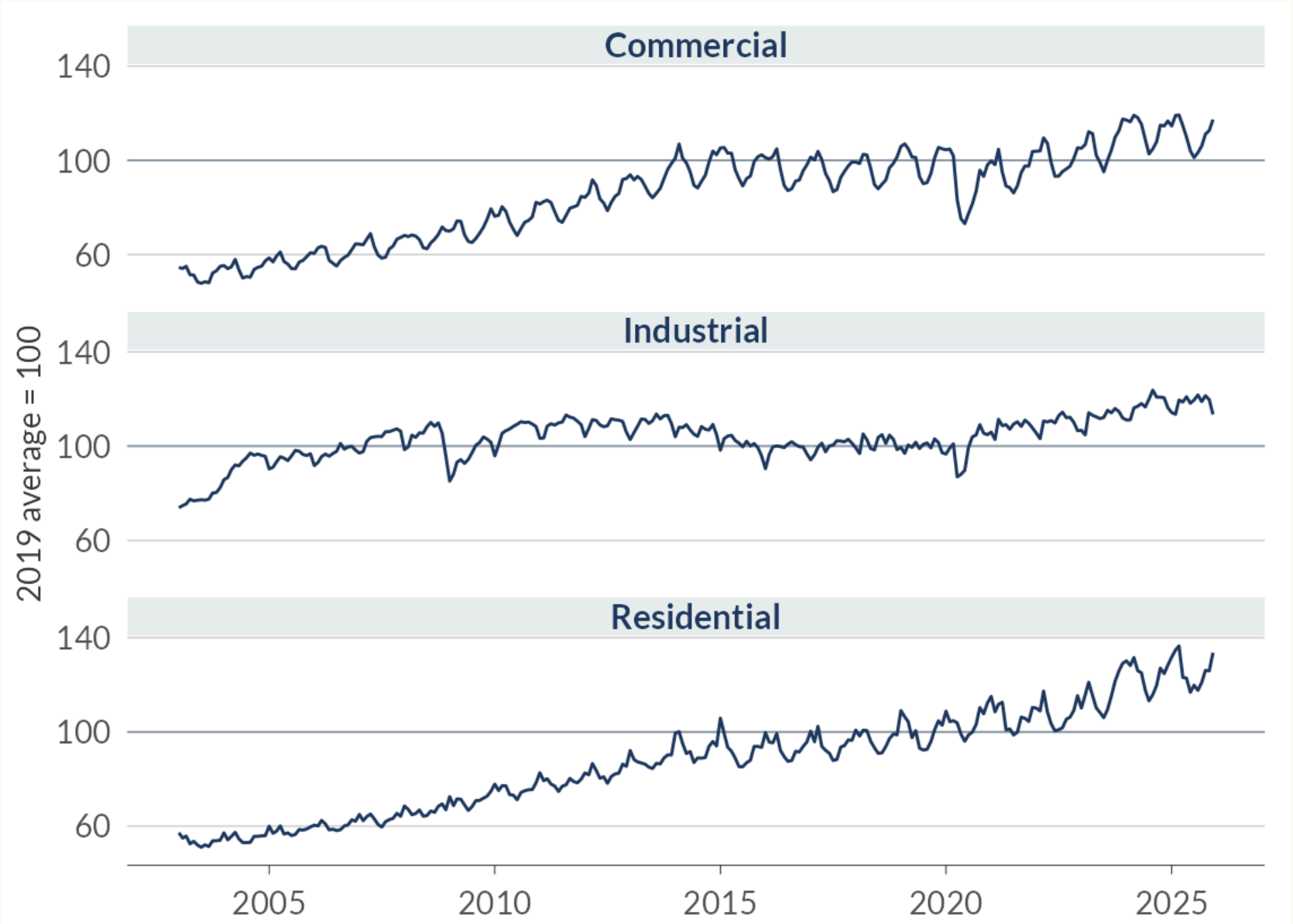
trendseries 1.7.0

Explore trends and cycles in R

Trends, cycles and rolling summaries in tidy tables.

Rebase many series to a common index

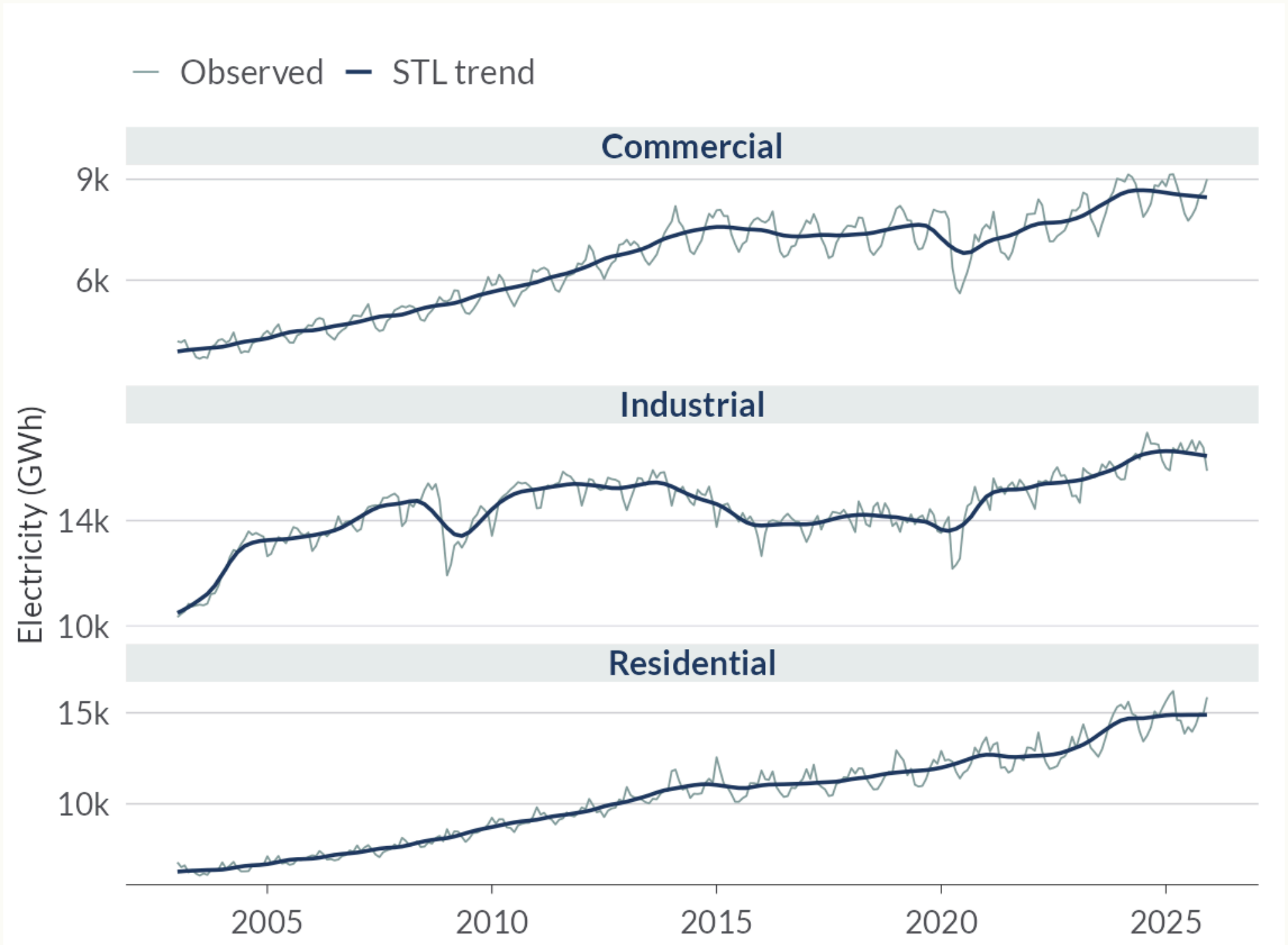



```
# rebase each series to its 2019 average
electricity |>
  dplyr::filter(date >= as.Date("2003-01-01")) |>
  index_series(
    group_cols = "name_series",
    base_period = 2019
  )
```



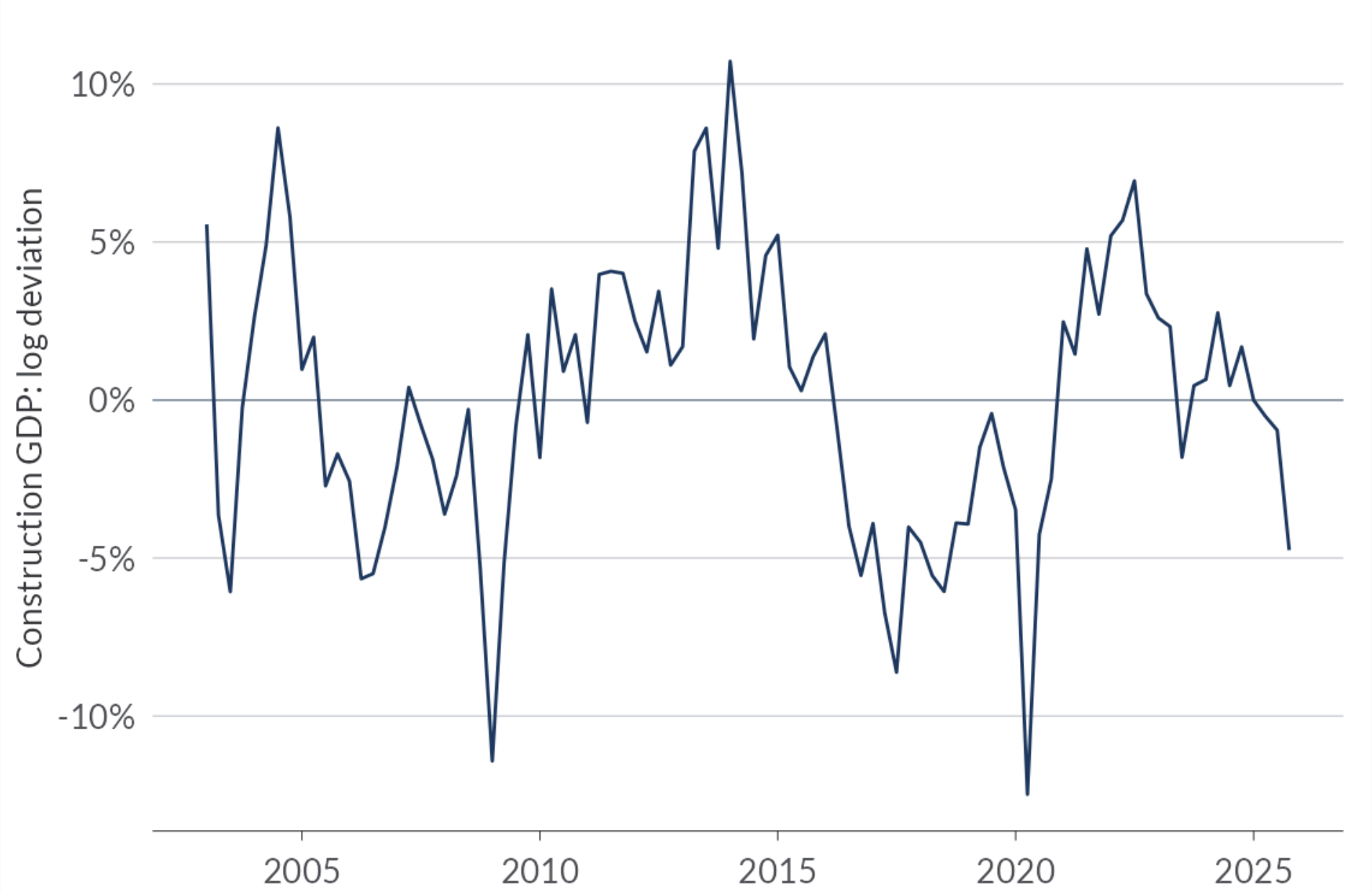
trendseries 1.7.0

Estimate trends for many series at once




```
# estimate an STL trend for each series
electricity |>
  dplyr::filter(date >= as.Date("2003-01-01")) |>
  augment_trends(
    group_cols = "name_series", methods = "stl"
  )
```

Separate trend and cycle in one pipe




```
# remove seasonality, then the trend (in logs)
gdp_construction |>
  dplyr::filter(date >= as.Date("2003-01-01")) |>
  deseason_series(value_col = "index") |>
  detrend_series(
    value_col = "seasadj_stl", transform = "log"
  )
```

> Log deviations approximate percentage deviations from trend.

New in 1.7.0

Use your tsibbles as they are

```
# estimate trends in a quarterly tsibble
gdp_construction |>
  dplyr::mutate(date = tsibble::yearquarter(date)) |>
  tsibble::as_tsibble(index = date) |>
  augment_trends(value_col = "index", methods = "hp") |>
  dplyr::slice_tail(n = 3)
```

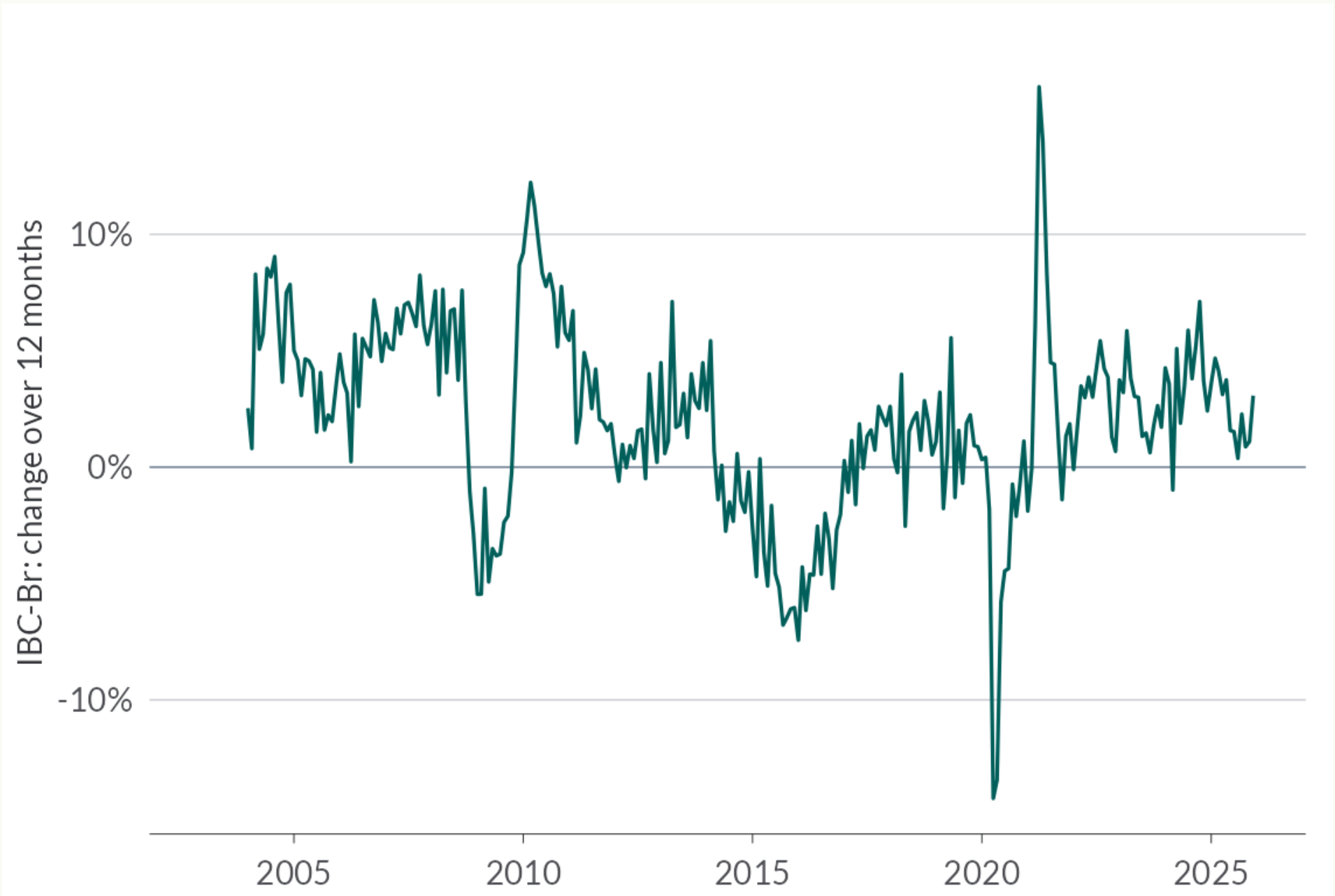
```
# A tsibble: 3 x 3 [1Q]
     date index trend_hp
    <qtr> <dbl>    <dbl>
1 2025 Q2  157.     160.
2 2025 Q3  164.     161.
3 2025 Q4  156.     162.
```

Construction GDP. The returned object keeps its tsibble class.

trendseries 1.7.0

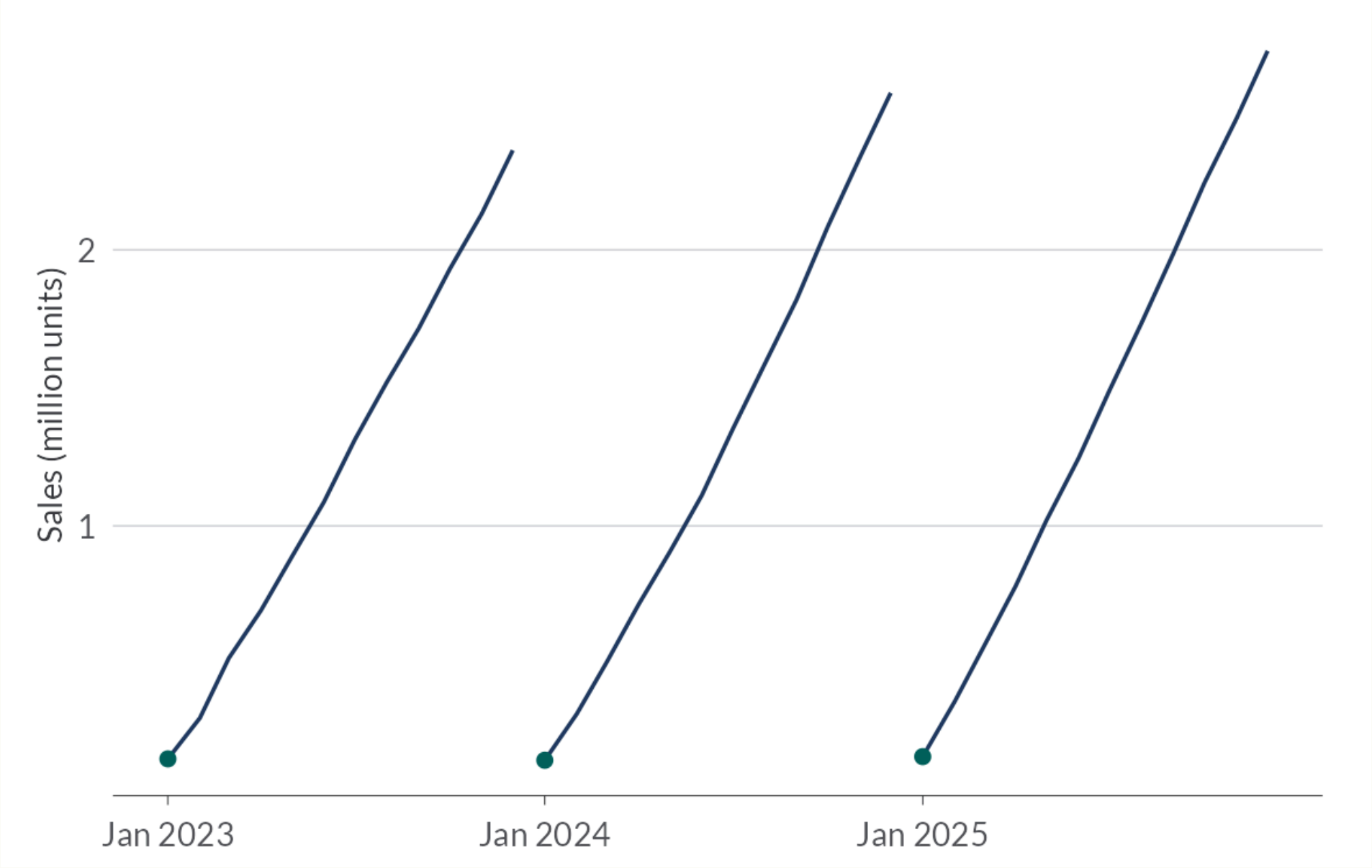
New in 1.7.0

Compute year-over-year change




```
# compare each month with a year earlier
ibcbr |>
  augment_rolling(
    value_col = "index", stats = "change",
    window = 12, percent = TRUE
  )
```

Accumulate values year to date




```
# sum sales from January to each month
vehicles |>
  dplyr::rename(sales = production) |>
  dplyr::filter(date >= as.Date("2023-01-01")) |>
  augment_rolling(
    value_col = "sales",
    stats = "sum", window = "ytd"
  )
```

> Dots mark the January resets.

Try it in R

Start with the bundled datasets. Find methods and examples in the docs.

```
install.packages("trendseries")
```

https://viniciusoike.github.io/trendseries/